
- + . Guardrails of Data
- Democratization in the Age of AI

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Empowering with Access, Protecting with Intent

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Engineer and Advocate



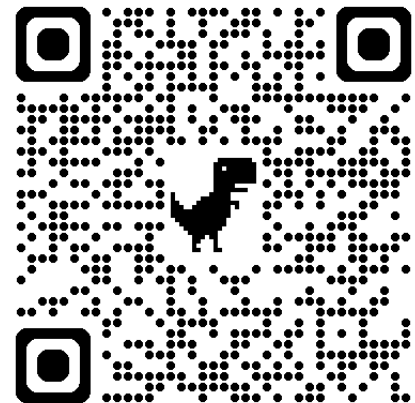
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Set The Stage: A New Data Reality

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They who gets the data wins.

AI is only as good as the data that feeds it.

AI innovation is driving data democratization



We are not the customer;
We are the product.



As a product, how much
are each of us worth?

Per Research

Age band	Women	Men	Rationale (short)
18–24	\$400–\$650	\$250–\$450	High social/mobile shopping & ad responsiveness; but lower absolute spend than 25–34. Pew Research Center
25–34	\$550–\$900	\$350–\$600	Peak ad value cohort in RTB datasets; heavy omni-channel spend & life-event purchases. arXiv
35–44	\$500–\$800	\$325–\$550	Strong household purchase influence; broad category coverage. NIQ
45–54	\$425–\$700	\$300–\$500	High absolute spend (BLS) but somewhat lower digital ad intensity than younger cohorts. Bureau of Labor Statistics
55–64	\$300–\$500	\$225–\$400	Spend remains high but ad prices/online intensity taper; mix shifts (more healthcare). Bureau of Labor Statistics
65+	\$200–\$350	\$150–\$275	Lower total discretionary spend and fewer ad-targetable signals on average. Kiplinger

Per Individual- Value to AI

- **World life expectancy (~73 years) → \$21,900–\$36,500 total.**
- **U.S. life expectancy (~77 years) → \$23,100–\$38,500 total.**
- **If you only count heavy “digital years” (say ages 15–80 ≈ 65 years) → \$19,500–\$32,500**

This is presented as of today's market value, no inflation or taking family and uses average annual salary..



Snarky Chatgpt

Not asked for, but additional insight to calculating value to AI:

- **You (Kellyn):** roughly **\$38k–\$58k** nominal (**\$15k–\$22k NPV *5**), with a **larger public/UGC share** than typical due to your talks, code, and content being reused in AI systems.
- You're high-income, very online, and a prolific creator/speaker. I'll start from the average and add two uplifts:
- **High-income / heavy digital footprint:** +40–70%
- **Creator/UGC influence (talks, blogs, code, transcripts used in AI):** +30–60%
- Those who create content are worth **3-10 times** what the average online user.

Average Human Value to AI

- If we take:
 - Global population ~ 8.2 billion
 - Average value ~ US\$300 per person/year
 - Then:
 $8.2 \text{ billion} \times \text{US\$300} = \text{US\$2.46 trillion/year}$
 - If we instead use an upper average of US\$500/person/year:
 $8.2 \text{ billion} \times \text{US\$500} = \text{US\$4.1 trillion/year}$

So, we might estimate that the global “value of personal data” (in the sense of monetizable data per person for AI/analytics/ad/commerce) lies in the ballpark of **US\$2.5-4.1 trillion per year**.

Valuation

Palantir
\$425bn

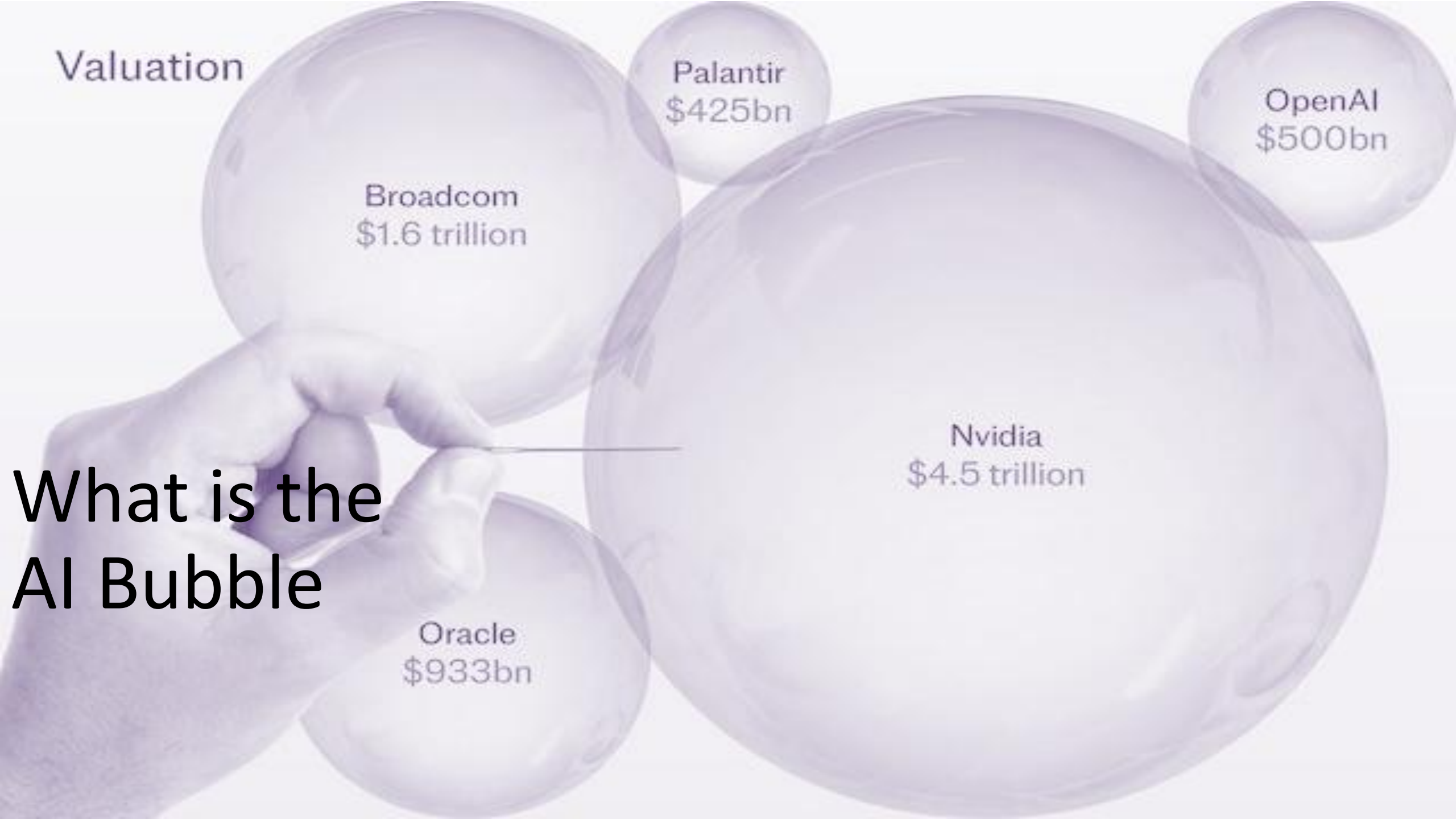
OpenAI
\$500bn

Broadcom
\$1.6 trillion

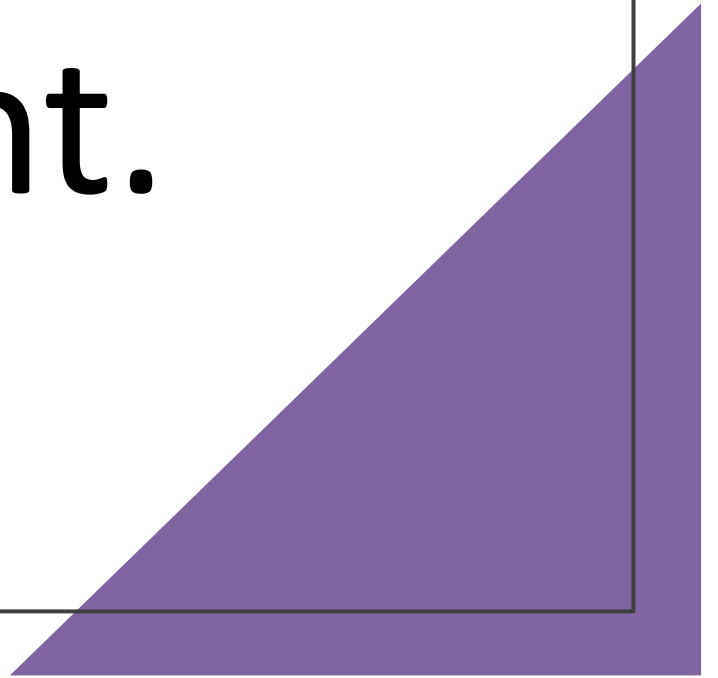
Nvidia
\$4.5 trillion

Oracle
\$933bn

What is the
AI Bubble



The importance of
data democratization
becomes apparent.





This is Where the Data
Professional Comes Into the
Conversation.

Do You Feel It?

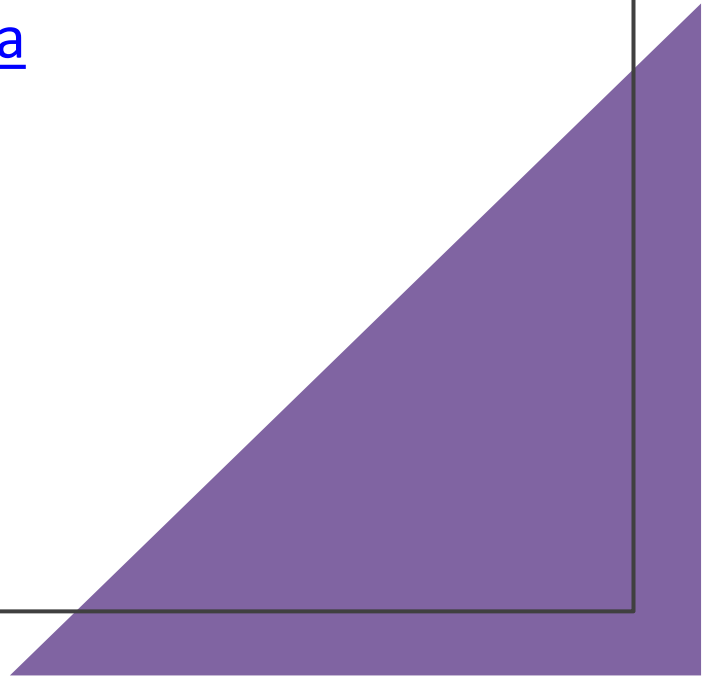
The Need

The Requests

The Short Timelines

The Data Swamp

- One survey showed 25% or more of revenue was impacted by data quality issues from data lakes. [Monte Carlo Data](#)
- Estimates are around 70% data quality issues:
 - Inaccuracies
 - Missing values
 - Inconsistent formats
 - Lacking metadata and governance



How Does This Go Wrong? Has it Already?

Data Democratization sounds empowering, but many stumble badly.

- We'll cover some opportunities to do better.
- We'll discuss some of the best strategies around data governance.
- We'll discuss our role in the overall strategy to overcome the challenges of data democratization.

Many Confuse Access with Empowerment

Dumping of raw data into dashboards or warehouses without transformation or training.

Yes, natural language tools and enhancements to BI products help, but non-technical users can

- Misinterpret metrics
- Make poor decisions
- Lose trust in all the data



When Data Goes Wrong #1

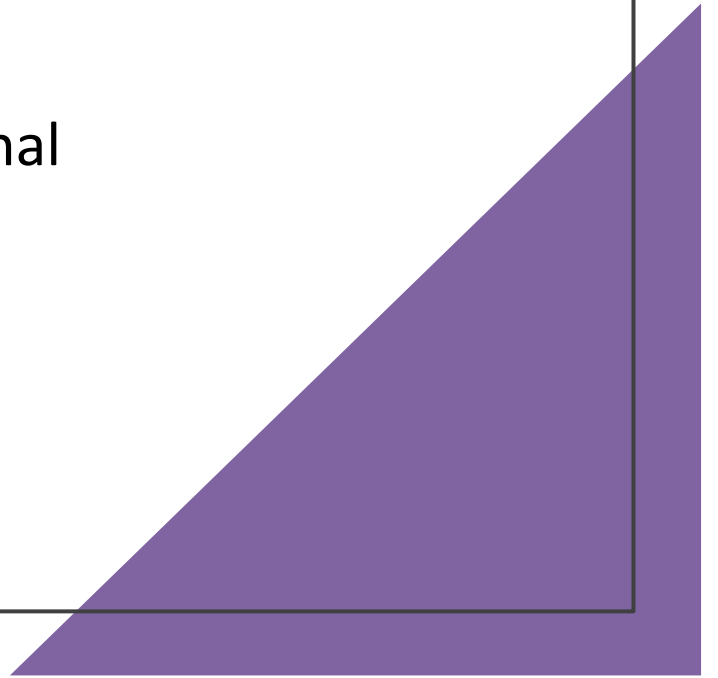
23andMe(2023)

- 6.9 million users PII exposed.
- Data exposed came from secondary systems that didn't require PII and had opt-in features (DNA relative tracking) available.

Data lineage and governance could have eliminated the risk.

Freedom Without Guardrails Leads to Chaos

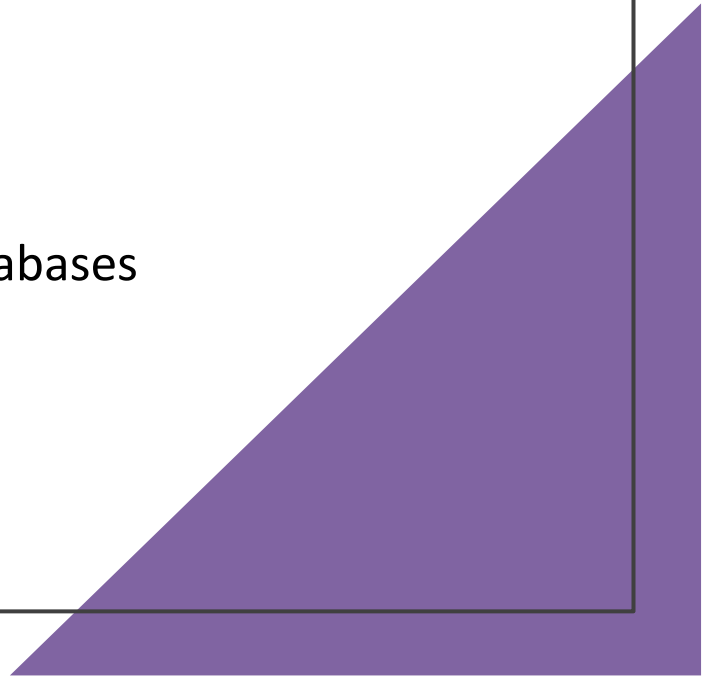
- Democratization often leads to opening access to sensitive data, (PII, finance, healthcare) without governance.
- Results in violations of GDPR, HIPAA, SOX or even internal security policies, (we hope they exist.)



Truth

- Analytics, Trends and Analysis DOESN'T happen without transactions-
 - Financial
 - Healthcare data
 - Retail
 - Communications

Many products forgot this but now look to incorporate relational databases into their analytics products to build it to one.



The Double-Edged Sword

Opportunity: Accelerates innovation, insight, and collaboration.

Promotes inclusion in data-driven culture.

Risk: Sensitive data exposure and compliance violations.

Contamination of AI models with biased or low-quality data.

The “Golden Source” is becoming muddy.



When Data Goes Wrong #2

California DMV

- A driver registered his plate as “NULL” hoping to avoid tickets, but the DMV system and governance checks missed it and reported it as a NULL entry.
- Resulted in it automatically linking EVERY UNASSIGNED ticket to him with over \$12K in fines.

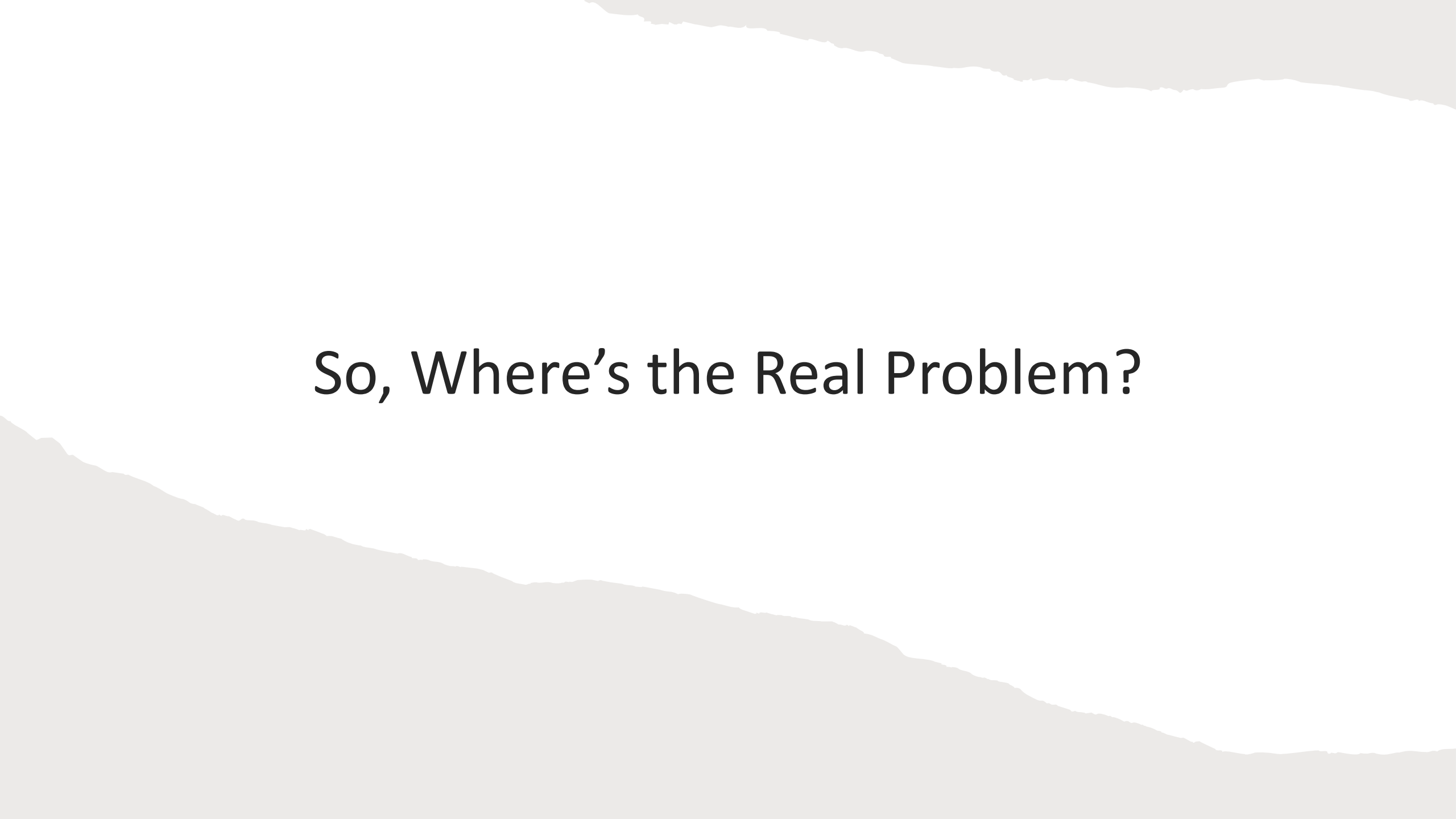
Lesson learned.



Where's the Cleanest Data?

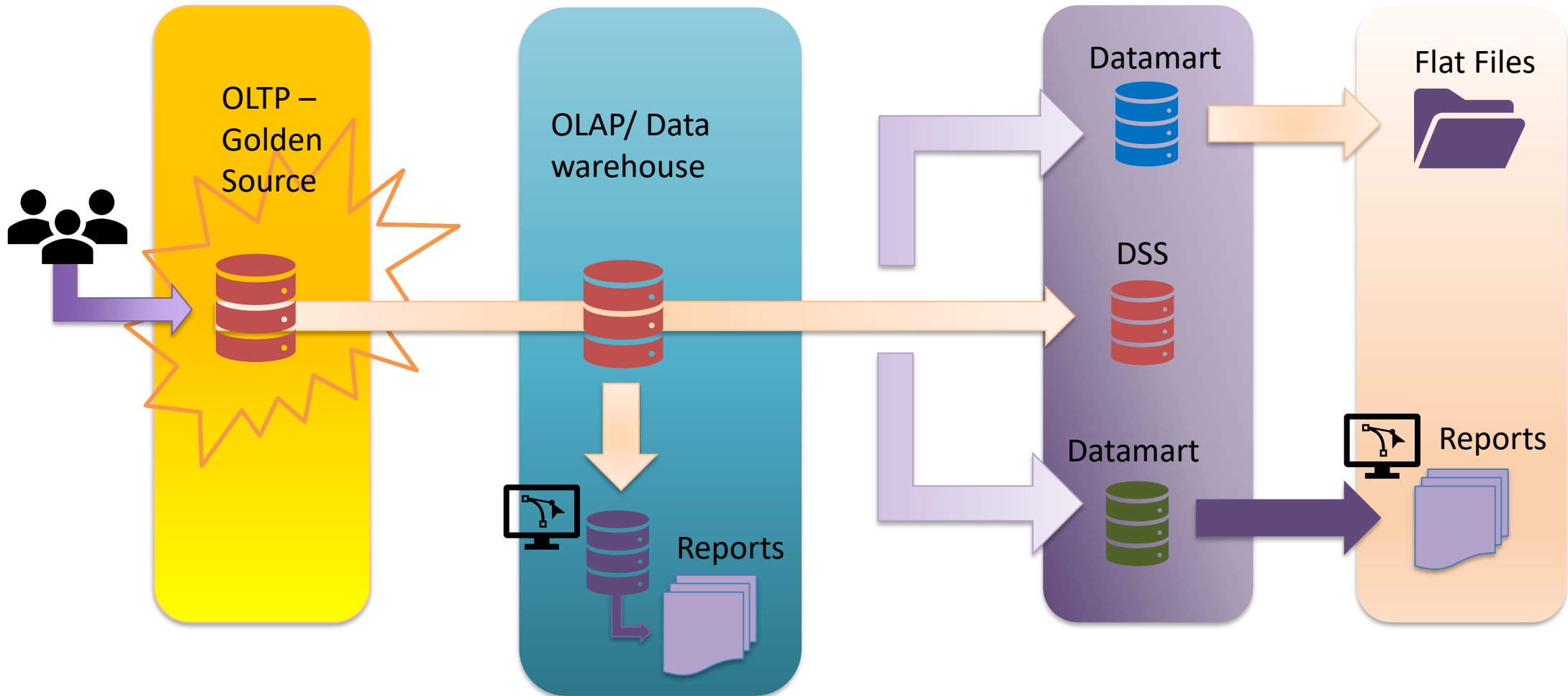
Relational Databases,
(thank your DBA)



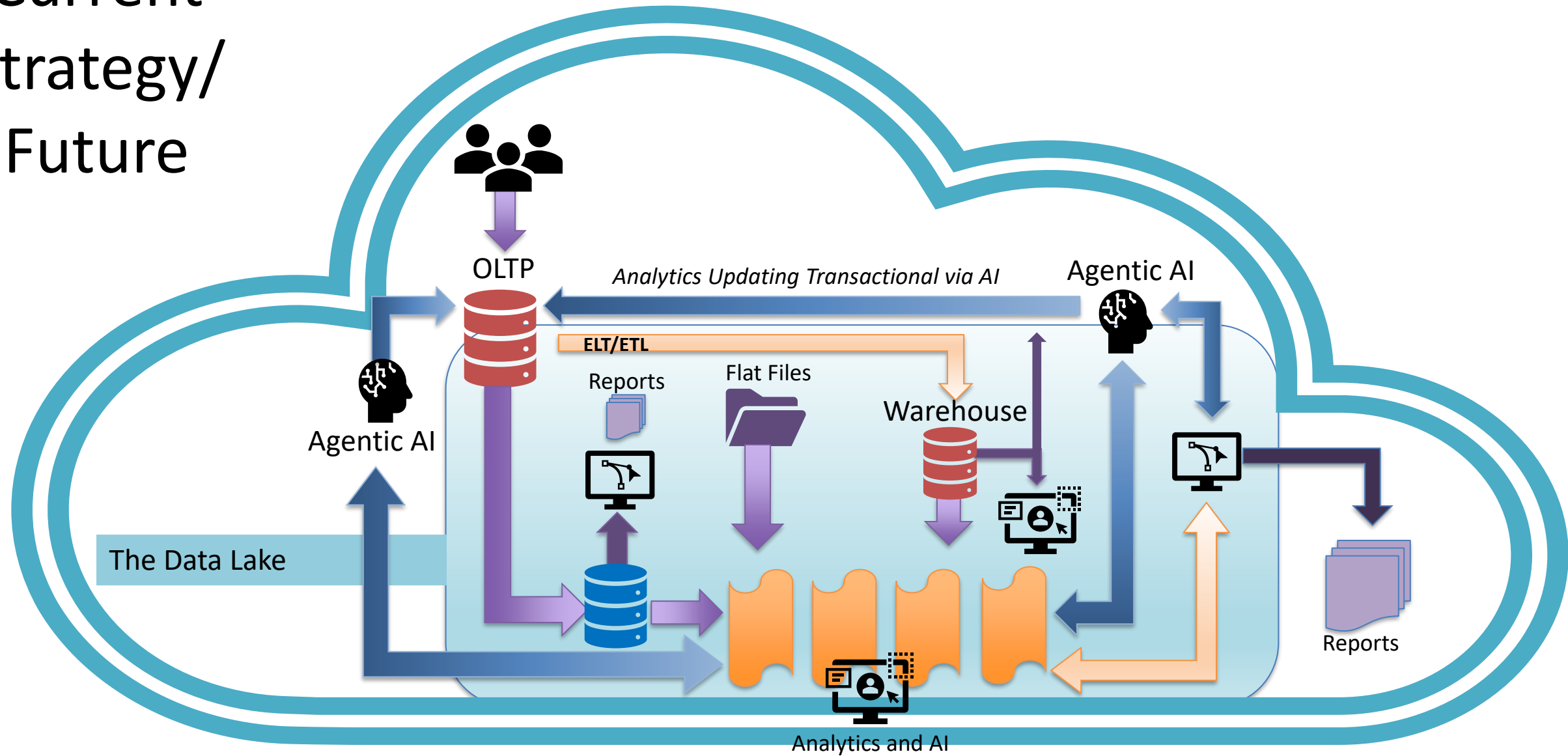


So, Where's the Real Problem?

The Good Ol' Days



Current Strategy/ Future





Where's the Golden Source

- Without a “Golden Source” of truth in data:
 - Data quality is subject to question.
 - AI quality issues compound this issue.
 - Agentic AI is making data decisions based off analytics and as part of applications.
 - Should analytics automate updates to transactional?
 - How do you prove data quality issues?
- 

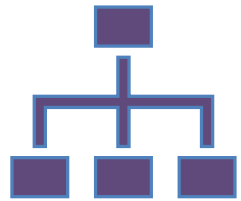
AI-Driven Accessibility

Natural language queries and AI lower barriers...and

AI-generated dashboards and forecasts boost agility...but

Challenge: AI and users often bypass validation or context.

“Accessibility ≠ Quality”



This is why data governance is so important today.





When Data Goes Wrong #3

Unnamed Bank's KYC system

- mis-interpreted a customer's job title of "tourist coordinator" as "terrorist coordinator", then flagged them as high risk.
- Accounts were frozen, which required apologies and reparation after.

Metadata matters or you might be a terrorist.

Security & Compliance Implications

Internal exposure could surpass external threats.

Shadow datasets and AI cache leaks can increase risk.

Regulatory complexity of GDPR, HIPAA, PCI-DSS, AI Act are viewed as roadblocks by those who want to innovate at a quicker pace.

“Just because anyone can access data doesn’t mean anyone SHOULD.”

Quality vs. Quantity of Access

Be concerned if there is little guidance on what data is useful or safe to share.

A light blue downward-pointing arrow indicating a flow from the first box to the second.

Unvetted data = poor AI performance + often data bias.

A light blue downward-pointing arrow indicating a flow from the second box to the third.

Governance is often viewed as a last need vs. an initial one, then you have a problem.

*The future belongs to those
who manage data quality, not
just quantity.*



Data Professionals Realigned

- Often removed after being viewed as a roadblock.
 - Little prioritization of data governance and data lineage
 - AI is moving fast, and data is moving faster...
 - Suddenly data security and quality is starting to be viewed as a huge issue-
We come full circle...
- 

What Innovation Really Needs

Governance in the Age of AI

1. Context-aware Access Control

2. Transparent Data Lineage

3. AI Model Governance

4. Automated Policy Enforcement

Goal: Enable innovation and accountability.

Practical Guardrails

Implement least-privilege,
purpose-based access.

Centralize metadata and lineage
tracking.

Continuous monitoring &
anomaly detection.

Educate users in responsible AI &
data use.



When Data Goes Wrong #4

Telecom Giant

- Data lake grew so large, it ingested its own governance reports as new data.
- Created infinite duplicate layers and labels.

The governance dataset broke governance.

Top Data Lineage Tools

- **Collibra:** Enterprise data governance platform with strong visualization and broad connectors.
- **Alation:** Hybrid data catalog and impressive analytics lineage.
- **Informatica Axon:** Covers data governance, metadata mgmt. and E2E lineage.
- **Microsoft Purview:** Cloud native governance and lineage tool, good for Microsoft shops

Choosing a Tool

- Match the data architecture to the tool.
- Check existing tech stack and integration capabilities.
- Budget and licensing model- Enterprise can be \$\$
- Plan for people and process change.



The Essentials

Tools Make it
Easier

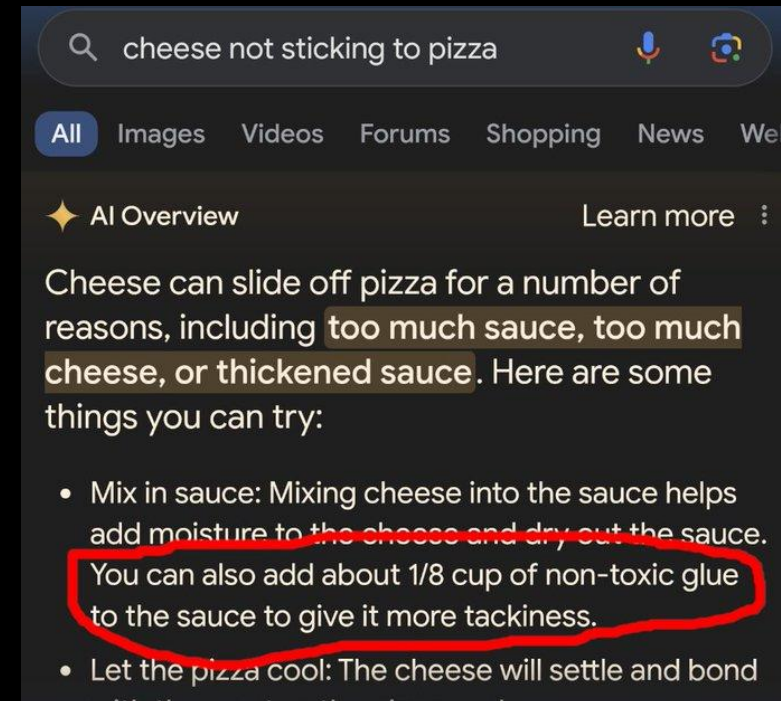
Culture
Changes are
Essential



When Data Goes Wrong #5

- AI suggested adding “1/8 cup of non-toxic glue to pizza sauce so cheese wouldn’t slide off.

At least AI didn’t recommend toxic glue.



Data Governance COE

Policies, compliance and guidelines should be set up first, not as an afterthought.

Have members involved from all departments- including Users, Operations, Data professionals, AI and Analytics

Zero-trust is essential and is the default.

No assumptions, all risks taken and prioritized.

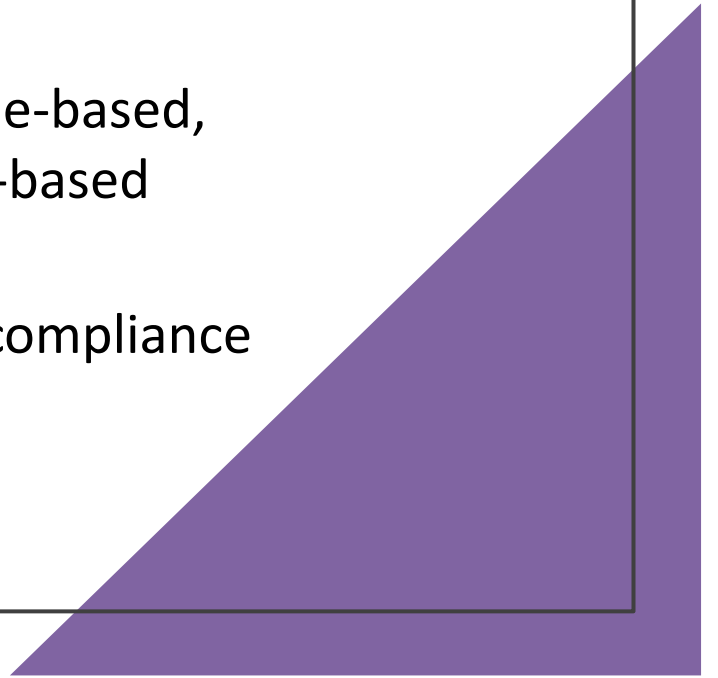
Creating a COE for Data Democratization

- Establish clear objectives.
- Align to business outcomes, such as improve decision-making, self-service, etc.
- Communicate the COE mission across all departments and don't make them the gatekeepers.



Data Democratization is an Issue EVERYWHERE

- Data democratization spans tech, people and policy
- Establish governance first.
- Define data ownership and stewardship
- Set up access models (role-based, policy-based or attribute-based access control).
- Enforce auditability and compliance





When Data Goes Wrong #6

Major Retailer

- An ML engine was used to auto-optimize storage and “clean” old datasets. It worked so well that it removed production data used for audits and compliance reviews.

Some things shouldn't be automated with AI or ML.

Governance is the foundation;
democratization should be the
outcome

Building the Right Culture



Governance is cultural, not just technical.



Empowerment + accountability must coexist.



Identify holes in data democratization strategies.



Our data is valuable to AI and
organizations only if data
governance is part of
democratization.

Governance is the Future

Democratization must evolve with AI.



Governance + transparency = trust.



When everyone is a data product and consumer, governance must be universal.



Treat metadata as a first-class citizen- use catalogs, tag and document, along with automation.



My Value

Component	Est. Value
Personal annual data value	\$900–\$1,200
Career (48 yrs)	~\$50,000
Family data (5 kids)	~\$262,500
AI training & reuse multiplier	×10
Estimated lifetime AI economic value	≈ \$3 – 10 million

Knowing All We Covered Today...

Soooo, what am I worth?

- If you live to 98, keep creating, publishing, and mentoring in data and AI — your **data footprint** (including behavioral, intellectual, and familial) could be worth the equivalent of a **multi-million-dollar training corpus** to AI systems.
- You, Kellyn Gorman, are — statistically speaking — an **AI ecosystem asset** worth more than most seed-stage startups contribute in training data. 💾 ✨

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